

FDS EXPERIMENT-13

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Experiment to handle missing values and inappropriate data in a given data set.

```
import pandas as pd
import numpy as np

data = {
    "Name": ["Arun", "Bala", "Charan", "Divya",
             "Ezhil", "Bala"],
    "Age": [20, 21, np.nan, 22, -5, 21],
    "Marks": [85, 90, 75, np.nan, 120, 90],
    "Department": ["CSE", "cse", "ECE",
                   "ECE", "CSE", "cse"]
}

df = pd.DataFrame(data)

print(df)
```

...	Name	Age	Marks	Department
0	Arun	20.0	85.0	CSE
1	Bala	21.0	90.0	cse
2	Charan	NaN	75.0	ECE
3	Divya	22.0	NaN	ECE
4	Ezhil	-5.0	120.0	CSE
5	Bala	21.0	90.0	cse

```

df["Age"] = df["Age"].fillna(df["Age"].median())

df["Marks"] = df["Marks"].fillna(
    df["Marks"].median()
)

print(df)

```

```

...      Name  Age  Marks Department
0     Arun  20.0   85.0          CSE
1     Bala  21.0   90.0          cse
2   Charan  21.0   75.0          ECE
3    Divya  22.0   90.0          ECE
4    Ezhil  -5.0  120.0          CSE
5     Bala  21.0   90.0          cse

```

```

df.loc[df["Age"] < 0, "Age"] = np.nan

df.loc[
    (df["Marks"] < 0) | (df["Marks"] > 100),
    "Marks"
] = np.nan

df["Age"] = df["Age"].fillna(df["Age"].median())
df["Marks"] = df["Marks"].fillna(df["Marks"].median())

print(df)

```

```

...      Name  Age  Marks Department
0     Arun  20.0   85.0          CSE
1     Bala  21.0   90.0          cse
2   Charan  21.0   75.0          ECE
3    Divya  22.0   90.0          ECE
4    Ezhil  21.0   90.0          CSE
5     Bala  21.0   90.0          cse

```

```
df["Department"] = df["Department"].str.upper()

print(df["Department"])
```

```
0    CSE
1    CSE
2    ECE
3    ECE
4    CSE
5    CSE
Name: Department, dtype: object
```

 `df = df.drop_duplicates()`

```
print(df)
print(df.isnull().sum())
print(df.duplicated().sum())
```

	Name	Age	Marks	Department
0	Arun	20.0	85.0	CSE
1	Bala	21.0	90.0	CSE
2	Charan	21.0	75.0	ECE
3	Divya	22.0	90.0	ECE
4	Ezhil	21.0	90.0	CSE

```
Name      0
Age        0
Marks      0
Department 0
dtype: int64
0
```